

ASSIGNMENT 8

1.WAP that reads text from a file and writes into another file.

```
try:

    with open('context.txt','r') as file:

        f = open('result.txt','a')

        for line in file:

            f.write(line.strip()+'\n')

        f.close()

except FileNotFoundError:

    print("file does not exist")

print('hello')
```

Python > Assignment >  context.txt

```
1  this is context file.
2  course - mca
3  college - igdtuw
```

Python > Assignment >  result.txt

```
1  this is context file.
2  course - mca
3  college - igdtuw
```

2.WAP that reads a file and print only those lines that has the word python

```
try:

    with open('myfile.txt','r') as file:

        for line in file:

            if 'python' in line.lower():

                print(line.strip())

except FileNotFoundError:

    print("file does not exist")
```

Python > Assignment > myfile.txt

```
1 I am learning Python to build web applications.
2 Python is widely used in data science and artificial intelligence.
3 Writing clean code in Python makes debugging much easier.
4 The teacher explained the concept with a real-life example.
5 She practices coding daily to improve her problem-solving skills.
```

PS C:\Users\sweta kumari\Desktop\python\assignment> python a8.py
I am learning Python to build web applications.
Python is widely used in data science and artificial intelligence.
Writing clean code in Python makes debugging much easier.

3. WAP to compare two files.

```
try:

    f1 = open("myfile.txt")
```

```

f2 = open("myfile2.txt")

f1_data = f1.readlines()

f2_data = f2.readlines()

i = 0

for line1, line2 in zip(f1_data, f2_data):

    i += 1

    if line1 == line2:

        print(f"Line {i}: IDENTICAL")

    else:

        print(f"Line {i}:")

        print(f"\tFile 1: {line1.strip()}")

        print(f"\tFile 2: {line2.strip()}")

f1.close()

f2.close()

except FileNotFoundError:

    print("file does not exist")

```

```

PS C:\Users\sweta kumari\Desktop\python\assignment> python a8.py
Line 1: IDENTICAL
Line 2: IDENTICAL
Line 3: IDENTICAL
Line 4: IDENTICAL
Line 5:
        File 1: She practices coding daily to improve her problem-solving skills.
        File 2: She practices coding daily.
_

```

4. WAP that accept a file name and a character as input from the user and count no. of times a character appear in a file

```
try:

    with open(filename, 'r') as file:

        count = 0

        data = file.read()

        for i in data:

            if i==ch:

                count += 1

        print(f"count = {count}")

except FileNotFoundError:

    print("file does not exist")
```

```
PS C:\Users\sweta kumari\Desktop\python\assignment> python a8.py
Enter your file name: demo.txt
Enter a character to search:l
count = 3
```

5. WAP to perform linear search and binary search.

```
# linear search
arr = [2,6,4,3,9,3,12]
num = int(input('Enter a number to search in list: '))
```

```
i = 0
flag = 0
for n in arr:
    if n == num:
        flag = 1
        break
    i += 1
if flag == 1:
    print(f'{num} is found at index {i}')
else:
    print('Not found')
#binary search
print("this is binary search")
numbers = [1,2,3,4,5,7,8]
num = int(input('Enter a number to search in list: '))
low = 0
high = len(numbers)-1
found = False

while (low<=high):
    mid = (low + high)//2
    if numbers[mid] == num:
        print(f'{num} is found at index {mid}')
        found = True
        break
    elif numbers[mid]<num:
        low = mid + 1
    else:
        high = mid - 1

if not found:
    print('Not found')
```

```
PS C:\Users\sweta kumari\Desktop\python\assignment> python a8.py
Enter a number to search in list: 5
Not found
PS C:\Users\sweta kumari\Desktop\python\assignment> python a8.py
Enter a number to search in list: 12
12 is found at index 6

PS C:\Users\sweta kumari\Desktop\python\assignment> python a8.py
this is binary search
Enter a number to search in list: 12
Not found
PS C:\Users\sweta kumari\Desktop\python\assignment> python a8.py
this is binary search
Enter a number to search in list: 5
5 is found at index 4
```

6. WAP to perform sorting in a list using sort and sorted.

```
numbers = [12, 4, 56, 17, 8, 99, 23]

print("Original List:", numbers)

numbers.sort()
print("List after sort():", numbers)

numbers = [12, 4, 56, 17, 8, 99, 23]

sorted_list = sorted(numbers)
print("List after sorted():", sorted_list)
print("Original List remains unchanged:", numbers)
```

```
PS C:\users\sweta kumari\desktop\python\assignment> python a8.py
Original List: [12, 4, 56, 17, 8, 99, 23]
List after sort(): [4, 8, 12, 17, 23, 56, 99]
List after sorted(): [4, 8, 12, 17, 23, 56, 99]
Original List remains unchanged: [12, 4, 56, 17, 8, 99, 23]
```